

Technology Stack

Date	2 July 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap	Used to develop a responsive and user-friendly web interface for login, flood prediction, prediction history, PDF download, and project information pages. Bootstrap ensures a clean and responsive design across different screen sizes.
2	Backend / Server-Side	Python, Flask	Python was used for implementing the machine learning logic, while Flask provides a lightweight web framework for handling routing, user authentication, prediction requests, and communication between the frontend and backend.
3	Database / Data Storage	SQLite, SQLAlchemy	SQLite stores user accounts and flood prediction history locally. SQLAlchemy simplifies database operations through an Object Relational Mapping (ORM) approach, making data management efficient and maintainable.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
4	Cloud / Hosting / Deployment	Local Flask Development Server (Future: IBM Cloud / Render)	The application is currently developed and tested locally using the Flask development server. Its modular architecture supports future deployment on cloud platforms such as IBM Cloud or Render for improved scalability and remote accessibility.
5	Version Control & CI/CD	Git, GitHub	Git is used for version control to track project changes, while GitHub provides secure repository hosting, collaboration, and source code management throughout the project development lifecycle.
6	Third-Party APIs / Other Tools	Scikit-learn, XG-Boost, Pandas, NumPy, Matplotlib, Seaborn, Flask-Login, Joblib, ReportLab, Visual Studio Code, Anaconda	These tools support data preprocessing, visualization, machine learning model development, user authentication, model serialization, PDF report generation, and the overall development environment used to build and test the application.